# GITAM Deemed to be University

Electronics and Communication Engineering | Course Code: EC305

## Electromagnetic Waves & Transmission Lines - Electromagnetic Field Theory &

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Overview: Wave propagation in lossy and lossless dielectrics, skin depth, Brewster angle, stub

## I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

**Unit 1: Maxwell Equations & Static Fields**
Gauss, Faraday, and Ampere laws in differential and integral forms, Displacement current, Boundary c

**Unit 2: Uniform Plane Wave Propagation**
Helmholtz wave equation, Propagation constant, Attenuation constant, Intrinsic impedance, Skin depth

**Unit 3: Reflection & Refraction at Boundaries**
Normal and oblique incidence, Snell law, Fresnel reflection and transmission coefficients, Total int

**Unit 4: Transmission Line Analysis & Smith Chart**
Line equations, Characteristic impedance, Reflection coefficient, VSWR, Single and double stub tunin

## II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Derive the wave equation in a conducting medium from Maxwell equations and obtain expressions for alph
2. Using a Smith chart, design a single-stub parallel tuner to match a load of Z_L = 100 + j80 ohms to a
3. State and prove Poynting Theorem for electromagnetic energy conservation in a finite volume.

## III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Matthew N.O. Sadiku - Elements of Electromagnetics Oxford University Press 7th Ed
* E.C. Jordan, K.G. Balmain - Electromagnetic Waves and Radiating Systems Prentice Hall

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